Clustering Assignment

Market Segmentation: Finding Structure in Customer Behavior

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Program	Fusemachines AI Fellowship 2026
Phase	Phase 2 : Foundational Statistical Learning
Submission	Jupyter Notebook (.ipynb) all cells executed

Context

A mid-sized retail company has transactional and behavioral data on approximately 5,000 customers spanning purchase history, frequency, recency, spend categories, and basic demographics. The marketing team wants to move away from broad campaigns and instead target distinct customer segments with personalized strategies. They have no labels — no one has ever segmented these customers before. Your job is to find the natural groupings in the data and tell a coherent story about who these customers are.

Dataset

UCI Online Retail II — real-world dataset containing approximately 500,000 raw transactions. You must construct the customer feature matrix yourself before any clustering. This is intentional friction that mirrors real-world data engineering.

Link: https://archive.ics.uci.edu/dataset/502/online+retail+ii

Tasks

Task 1 — Feature Engineering

Your first job is to transform 500,000+ raw transactions into a single row per customer. Real-world data never comes pre-packaged for machine learning.

Start by cleaning the data: remove rows with missing CustomerIDs, cancel orders (invoices starting with 'C'), and any rows with zero or negative quantities or prices.

Minimum required features (RFM):

- **Recency** — number of days since the customer last purchased, measured from the dataset's most recent date.
- **Frequency** — number of unique invoices for that customer.
- **Monetary** — total spend calculated as quantity multiplied by unit price.

Extended features (push further):

- Category spend ratios — what percentage of each customer's total spend falls in each product category.
- Return rate — ratio of cancelled to total quantity per customer.
- Average basket size — total revenue divided by number of invoices.
- Average gap in days between purchases.

> *Each extra feature you engineer must be justified — explain why you believe it will help separate customers into meaningful groups.*

Task 2 — Preprocessing

Before clustering, your data must be clean and scaled. For each preprocessing decision, document your reasoning explicitly.

- **Missing values** — decide whether to impute or drop. Document your reasoning.
- **Outliers** — identify using IQR or Z-score. Decide whether to cap, remove, or keep each one and justify.
- **Scaling** — apply StandardScaler or RobustScaler. Explain why you chose one over the other and why scaling matters specifically for distance-based algorithms.

Task 3 — K-Means Clustering

Run K-Means with at least two initialization strategies: k-means++ and random. Test values of k from 2 to 10.

- **Elbow Curve** — plot inertia (within-cluster sum of squares) against k. Look for the point where adding more clusters stops giving meaningful improvement.
- **Silhouette Score** — plot for each k. Scores closer to 1 indicate better-separated clusters.
- **Initialization comparison** — document whether k-means++ and random init gave different results. Compare inertia values and cluster assignments. Explain what this tells you about solution stability.

> *Choose your final k based on both plots together, not just one. Justify your decision.*

Task 4 — Hierarchical Clustering

Apply Agglomerative Clustering using at least two linkage methods. Plot a dendrogram for each. Choose your cut point where there is the largest vertical gap before the next merge — this indicates the clusters were very different before being joined. Justify your cut point explicitly.

Linkage	Description
Ward	Minimizes within-cluster variance. Tends to produce compact, evenly-sized clusters.
Complete	Uses maximum distance between clusters. Tends to produce tighter clusters.
Single	Uses minimum distance. Sensitive to outliers — good to understand but rarely best.

Task 5 — DBSCAN

DBSCAN groups points that are densely packed together and labels sparse points as noise. It requires two parameters: epsilon (neighborhood radius) and min-samples (minimum points to form a dense region).

- **Epsilon estimation** — plot the k-distance graph by computing the distance from each point to its k-th nearest neighbor and sorting these distances. The elbow suggests a good epsilon value.
- **Experimentation** — try at least three combinations of epsilon and min-samples. Build a comparison table.
- **Noise analysis** — analyze points labeled -1 carefully. Are they truly irrelevant outliers, or high-value anomalies like VIP customers or extreme one-time buyers? This analysis carries significant weight.

Task 6 — Cluster Validation

Compare the quality of your clustering results across all three methods using at least two metrics. Build a comparison table listing each method, number of clusters, and all computed metrics.

Metric	What it measures	Better when
Silhouette Score	How similar a point is to its own cluster vs other clusters. Range: -1 to 1.	Higher (closer to 1)
Davies-Bouldin Index	Average similarity between each cluster and its most similar neighbour.	Lower (closer to 0)
Calinski-Harabasz Index	Ratio of between-cluster variance to within-cluster variance.	Higher

For DBSCAN: exclude noise points (label = -1) before computing validation metrics. Explain why in your notebook.

Task 7 — Business Narrative

For each cluster identified by your best-performing method, write a short profile answering four questions:

- Who are these customers in plain language?
- What do their RFM and behavioral patterns look like?
- What business problem or opportunity does this segment represent?
- What specific marketing action do you recommend?

Example	RFM Pattern	Segment Name	Recommended Action
High-value loyals	Low recency, high frequency, high monetary	Champion Customers	VIP rewards or early product access
Lapsed customers	High recency, low frequency, low monetary	At-Risk / Lapsed	Win-back campaign with discount offer

Every recommendation must connect directly to the cluster's actual feature values. Do not write generic marketing advice.

Submission Requirements

Item	Requirement
File format	Single Jupyter Notebook (.ipynb) with all cells executed and no errors.
Code quality	All code blocks clearly commented. Comments must explain WHY, not just WHAT.
Visualisations	All plots labelled with titles, axis labels, and legends where applicable.
Business narrative	Written section (200-300 words) on cluster interpretation — readable by a non-technical stakeholder.
Failure log	Minimum 3 entries. Each entry: hypothesis tested, what happened, what you learned.

Honor Code

AI tools and LLMs are allowed for learning, brainstorming, debugging, and prototyping. However, they are tools to assist, not replace your critical thinking. Your submissions must reflect your own understanding.